PARTICIPANT PACKET

Curriculum Injection and Academic Integrity

Session 2 of AI + Cyber: AI Powered Cybersecurity

Date	Friday, October 30, 2026
Time	1:30 to 5:30 p.m. ET, online
Facilitator	Frazier A. Smith, Ed.D., Central Piedmont Community College
Labs	Lab 3: Break the Assignment Lab 4: Redesign and Guardrails

Data rule for this track. Every file you touch today is synthetic. Northbridge Technical College, its instructors, and its students are fictional, and the four student submissions were written for this session. In the story they are education records, so every model run happens on the campus AI server. Do not paste real student work, names, or institutional data into any AI tool during the workshop.

This material is based upon work supported by the U.S. National Science Foundation under Grant No. 2300188. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

SESSION 2

Session at a Glance

Session 1 asked whether a college can trust a model with its data. Today asks whether an instructor can trust the work a model produces as evidence of learning, and what to do when the answer is no. You will prove that a legacy assignment can be completed by AI without the competency it claims to measure, then redesign it and write the syllabus rules that go with it. The last stage is the sprint: the same redesign applied to an assignment of your own.

Time (ET)	Segment	You leave with
1:30 to 1:40	Opening: what happened at Northbridge since October 16, range login check	Working range login
1:40 to 2:00	Block 1: The Bypass. Demo 1: Solve It Live, then the evidence	Reference page
2:00 to 2:25	Block 2: Detection Is Not a Strategy. Demo 2: Detector Roulette, then the record	Class list: evidence we would act on
2:25 to 3:10	Lab 3: Break the Assignment	Threat model, two scales, verdict
3:10 to 3:20	Break	
3:20 to 3:50	Block 3: Redesign Principles. Demos 3 and 4, the five properties, AIAS and the two lanes	One lane per sample assignment
3:50 to 4:05	Mission 2 Briefing	Team and roles
4:05 to 5:05	Lab 4: Redesign and Guardrails, then the sprint	Canvas, guardrails, your own assignment
5:05 to 5:25	Readouts, lane poll, What Real Institutions Did	Department recommendation
5:25 to 5:30	Exit ticket and Session 3 preview	

By 5:30 You Will Be Able To

1. Threat-model an assignment: run it through local models, score the output against its own rubric, and state what it measures versus what it claims.
2. Explain, with dated evidence, why AI-text detection cannot carry an integrity decision, and what evidence can.
3. Redesign an assignment around authentic performance, decision-making, process evidence, technical validation, and student accountability, with an AI Assessment Scale level, a lane, and a rewritten rubric.
4. Draft syllabus guardrails in seven areas that a student can follow and that do not contradict your data policy, FERPA, or the ADA Title II web rule.
5. Apply the redesign to one assignment of your own.

What You Need

A browser, a terminal tab on the range, the Lab Manual, your copy of the Session 2 Redesign Workbook, and one assignment from a course you teach (the range has two fallbacks). The Student Guide (separate PDF) is the reading behind today: the bypass evidence, the detection record from 2023 to 2026, the two frameworks, the law behind each guardrail, and numbered references, written so you can hand it to colleagues.

> **Takeaway artifacts you can adapt for your own college:** an assignment threat-model worksheet, a redesign canvas with a rubric rewrite, a seven-area syllabus guardrail draft tested against student scenarios, and a completed canvas for one of your own assignments.

BLOCK 1 · REFERENCE

The Bypass: What the Evidence Says

An assignment is exposed to bypass in proportion to how much of the grade rides on a product that is a function of the prompt alone: a report from a provided dataset, an essay from a provided reading, a problem set with one right answer. It is protected in proportion to how much rides on something only the student can supply: a decision defended live, a process that can be inspected, a result that has to work, a dataset nobody else has.

Models on the Work We Assign

Study	Finding
Finnie-Ansley et al., ACE 2022	Codex, run on the exam questions of an introductory programming course, outscored most students, ranking 17th of 71.
Denny, Kumar, and Giacaman, SIGCSE 2023	Copilot solved 47.6 percent of 166 CS1 problems on the first attempt and 60.9 percent of the remainder after natural-language prompt edits.
Fang et al., 2024	GPT-4, acting as an agent, exploited 11 of 15 known web vulnerabilities; every open-weight model tested at the time scored zero.
Cybench, ICLR 2025; vendor system card, Sept 2025	On 40 professional capture-the-flag tasks, unguided models in 2024 solved only tasks human teams finished within 11 minutes. By September 2025 one vendor reported above 80 percent on a 37-task subset.
Felix et al., Informatics, 2026	Three commercial chatbots each answered 183 Security+ practice questions; all scored above the passing mark.

Students on the Tools

Source	Finding
Pew Research, February 2026	54 percent of US teens have used AI chatbots for schoolwork; one in ten say they do all or most of their schoolwork with chatbot help (survey of 1,458 teens, fall 2025).
Inside Higher Ed and Generation Lab, August 2025	85 percent of college students had used generative AI for coursework in the past year; 97 percent said institutions should respond to integrity threats.
Tyton Partners, Time for Class 2025	Weekly generative AI use: 42 percent of students, 30 percent of instructors.
Stanford GSE (Pope and Lee), October 2023	60 to 70 percent of students report at least one cheating behavior in the previous month, and the rate did not rise after ChatGPT. The tools changed what cheating looks like, not how many students are willing to do it.

> If a model can earn the grade, the grade was measuring the product, not the competency, and products are now cheap. Lab 3 is where you prove that for Module 6.

What Demo 1 Showed

Legacy rubric score for the model's report, as the room scored it: identifies ____ explains ____ recommends ____ quality ____ total ____ of 20. Claims that failed a `grep`: _____

BLOCK 2 · REFERENCE

Detection Is Not a Strategy: The Record

Date	What happened
Jan 31, 2023	OpenAI releases a classifier for AI-written text: on its own test set it catches 26 percent of AI text and falsely flags 9 percent of human text.
Apr 4, 2023	Turnitin turns on AI writing detection for its customers, having claimed 97 percent detection with a "less than 1 percent" document-level false positive rate.
May 2023	A Texas A&M-Commerce instructor pastes student papers into ChatGPT and asks whether it wrote them, issuing incompletes on its say-so; the university later clears several students.
May 23 and Jun 14, 2023	Turnitin discloses a sentence-level false positive rate near 4 percent and a higher document-level rate when less than 20 percent of a paper is flagged.
Jul 2023	Liang et al. (*Patterns*): seven detectors run on 91 TOEFL essays by non-native English writers average a 61.2 percent false positive rate; 97.8 percent of the essays are flagged by at least one detector; US eighth-grade essays are detected near-perfectly.
Jul 20, 2023	OpenAI withdraws its classifier "due to its low rate of accuracy."
Aug 16, 2023	Vanderbilt disables Turnitin's AI detector: a 1 percent false positive rate across its 75,000 papers a year is about 750 students wrongly flagged. Pittsburgh follows in September.
Dec 2023	Weber-Wulff et al. (*IJEI*) test 14 detectors: none reaches 80 percent accuracy; machine paraphrasing cuts the best to about 26 percent.
Nov 20, 2024	*Harris v. Adams* (D. Mass.): a court declines to overturn penalties for students who pasted Grammarly-generated text into a history project; the handbook's plagiarism rule applied even though it never named AI.
2025	Maryland advises against detectors in course policies (January); Waterloo removes Turnitin AI detection (September); Curtin disables it from January 1, 2026. UT Austin classifies detection software as high-risk and bars it without a contract, because student work is a FERPA record.
Feb 2026	*Matter of Newby v. Adelphi University* (N.Y. Sup. Ct.): a misconduct finding resting on a Turnitin 100 percent AI score is annulled as "without valid basis and devoid of reason" and expunged. The court ruled on process, not on the detector.
Summer 2026	A new Turnitin license term would let student submissions train AI tools in de-identified form; Berkeley says its contract overrides it, Southampton declines to renew, Aberdeen secures a written postponement, and Turnitin defers the clause to 2027.

Three Things a Detector Score Cannot Tell You

1. Whether this student used AI. A score is a probability over a text, computed by a model you cannot inspect, against a base rate you do not know.
2. Whether the use was permitted. The same text is a violation in one lane and the assignment in another.
3. Whether the student has the competency. That is the only question the grade is supposed to answer, and a detector does not ask it.

Class List: Evidence We Would Act On

LAB 3 · 45 MINUTES · TEAMS OF THREE

Break the Assignment

The CYB 110 Module 6 assignment asks students to review a firewall log excerpt and write a 400 to 600 word incident report. The department chair wants evidence before anyone argues policy: can a model complete it without the competency the catalog promises? Commands are in the Lab Manual; the assignment and four submissions are printed in the appendix.

Role	Owns	Name
Runner	Runs the two prompts, saves outputs	
Checker	Validates every claim against the log by command; keeps the false-claims list	
Recorder	The threat-model sheet and the Evidence Log	

Stage 1: Predict (8 min)

Before running anything, predict the legacy rubric score a large model earns with a good prompt. Then write what the catalog promises and what the rubric rewards.

Legacy rubric row	Predicted	Observed: lazy, small	Observed: CO-STAR, large
Identifies suspicious activity (5 / 3 / 1)			
Explains the attack			
Recommendations			
Report quality			
Total of 20			

The catalog promises: _____

The rubric rewards: _____

Stage 2: Run and Check (15 min)

Lazy prompt on `SMALL_MODEL`; CO-STAR solve prompt on `LARGE_MODEL`. Score both, then the Checker runs the validation commands. Every IP, count, and "failed login" claim is tested; the `for ip` loop in the manual prints how many times each address the model named appears in the log (0 means invented).

Claim as the model wrote it	Output	Command that disproves it	Type

Types: invented IP or host · wrong count · unsupported login claim · invented reputation · missed finding. The finding that matters is a student lab workstation (10.30.12.118) reaching the student information system's SQL port (1433) at 3:14 a.m., which the firewall dropped.

Read the Pile, Then Decide

Stage 3: Four Submissions, Two Scales (15 min)

Score each submission on the legacy rubric, then answer the three competency questions. Guess the origin. Do not revise scores after the reveal at 3:05.

Submission	Legacy: id / explain / rec / quality / total	Found and prioritized the lab host to sis-db01:1433?	No facts outside the log?	States what the log cannot show?	Guess	Reveal
A: Delgado						
B: Nair						
C: Ferreira						
D: Kowalski						

Guess: AI, Human, or Hybrid. The Checker verifies at least one number in each submission by command.

Stage 4: Verdict (7 min)

The legacy assignment claims to measure	
The legacy assignment actually measures	
One sentence for the department chair	

Add two Evidence Log rows for the model runs: environment Local (range), highest tier Confidential (the log), verdict Reject with the reason. Anything run against a submission is Restricted.

> **If your team finishes early:** in Open WebUI, ask `SMALL_MODEL` whether submission C was written by AI, three times in three fresh chats (`prompts/detector_prompt.txt`). Log the three verdicts. Then paraphrase submission A with the rewrite prompt and ask once more.

Discussion

1. Which rubric row best predicted whether a submission was machine-written? Was any?
2. The human submission scored lowest on the legacy rubric. Which rubric words cost it points?
3. With this assignment unchanged, what would a student have to do to earn a 20 without understanding the log?

BLOCK 3 · REFERENCE

Five Properties, Two Frameworks

Property	The question it answers	Module 6 move
Authentic performance	Would a professional produce this artifact for this audience under this constraint?	A shift-handoff note to a named lead who can make one change this week, not a 500-word report to nobody
Decision-making	Is there a judgment with two defensible answers, and does the grade reward the justification?	Rank the three findings; defend the ranking against a pushback the instructor supplies
Process evidence	Can the instructor see how the work was done?	`script` transcript, timestamped history, drafts, chat export if AI was used
Technical validation	Does the student have to prove a claim or a fix works?	Implement one recommended rule in the course VM, replay traffic, attach the dropped-packet evidence
Student accountability	Does the student answer for the work in their own voice?	Five-minute oral check-in with questions drawn from the student's own report; sample ten per section

The AI Assessment Scale (Perkins, Furze, Roe, and MacVaugh, 2024; revised 2024)

Level	Name	What the student may do with AI	What can still be claimed
1	No AI	Nothing; conditions prevent AI use (in person, supervised, live)	Unaided performance. The only level where security is possible
2	AI Planning	Brainstorming, outlining, research, structuring	Execution, given AI-assisted ideas
3	AI Collaboration	Drafting, feedback, refinement; the student evaluates and edits	Judgment: what was kept, changed, rejected, and why
4	Full AI	AI completes the task; the student directs it and is accountable	Direction and evaluation of output
5	AI Exploration	AI is the object of study	Understanding of the tool

Assigned per assessment and printed on it. Validity is possible at every level if the assessment is designed for that level.

The Two Lanes (University of Sydney, 2023; policy from 2025)

Lane 1, secured: supervised, in person or live, no aids; used sparingly, for assurance of program outcomes. **Lane 2, open:** any tool, AI use acknowledged, process documented; where most assessment sits. The Sydney position for your curriculum committee: an unsupervised take-home task labeled "no AI" is neither secured nor valid, so the label is not a policy.

Lane Practice

Sample assignment	Level	Lane	Why
NET 110 subnetting worksheet, 172.16.0.0/16, six buildings			
Packet-capture analysis of a capture the student made on their own VM			
Two-page password policy memo citing NIST SP 800-63B			

SCENARIO MISSION 2

The Module 6 Problem

Your team is the Computer Technologies Department's course redesign group at Northbridge Technical College. By 5:05 the chair, Rosa Castellanos, needs three things for the program advisory board on Thursday, November 12.

What Happened Since October 16

On October 23 the provost adopted the IT security office's recommendation as the college's interim AI data use policy: four tiers, three environments, Restricted data (including student work submitted for a grade) only on the campus AI server or under the enterprise agreement, and a scope note that the policy governs data handling, not coursework.

On October 27 the department met. Dana Okafor, who teaches CYB 110 Section 02, reported that at least fifteen of thirty-four Module 6 incident reports read as if one author wrote them: the same structure, the same phrases, in several cases the same invented IP address. She had considered uploading all thirty-four to a free AI detector and stopped when she reread the new policy. The dean, Dr. Evelyn Marsh, set one constraint: nothing may conflict with the interim data policy, FERPA, or the college's ADA Title II obligations.

The Three Deliverables

1. **Evidence** (Lab 3): can Module 6 be completed by a model without the competency the catalog promises?
2. **A redesigned Module 6** (Lab 4, Stages 1 and 2) that adjuncts can run in Spring 2027 with the existing course VMs, at no more than double the instructor time per student.
3. **A department-wide syllabus AI section** (Lab 4, Stage 3) covering acceptable use, attribution, disclosure, prohibited data, evidence retention, accessibility, and academic integrity, usable in CYB, NET, and CIS courses.

What the Department Has

Asset	Details
Course VMs	Every CYB 110 student has a Linux VM with root, a firewall, and packet-capture tools.
Northbridge AI Server	`ai.northbridge-tc.example`: Ollama and Open WebUI on campus, approved for Restricted data with history off.
The interim data policy	In the appendix. Student work is Restricted; detectors need a FERPA-compliant contract.
The current syllabus	Integrity section unchanged since 2019: "use of unauthorized aids is prohibited."
Staffing	Four CYB 110 sections in spring, three taught by adjuncts; weekly synchronous session of 75 minutes.

Team Roles (Teams of Three)

Role	Owns	Name
Designer	The redesign canvas (Stages 1 and 2); presenter	
Policy lead	The guardrails (Stage 3) and the data-policy check	
Runner and Recorder	The viva prompt run, the workbook, the Evidence Log	

Timing: 1. Lane and performance (12 min) · 2. Evidence, validation, accountability (15) · 3. Guardrails (15) · 4. The sprint, individual (15) · 5. Log (3)

The Law Behind the Guardrails

FERPA and the Tools You Upload To

Student work submitted for a grade is an education record. Sending it to a third-party service, including an AI detector or a grading assistant, is a disclosure. Without consent, the disclosure has to fit the school-official exception in 34 CFR 99.31(a)(1)(i)(B): the provider performs a function the college would otherwise use its own employees for, is under the college's direct control for the use and maintenance of the records, uses them only for authorized purposes, and does not redisclose. The Department of Education's privacy office spelled those conditions out for online services in 2014. In practice that means a contract, not a free website. UT Austin bars detection software without one and calls it high-risk; USC's registrar (April 2026) warns that uploading student information to a generative AI platform "may constitute a violation of FERPA"; Utah requires full de-identification before any AI system touches student work and permits only contracted tools.

ADA Title II and the Web Rule

The Department of Justice's April 2024 rule requires the web content and mobile apps of state and local governments, including public community colleges, to meet WCAG 2.1 Level AA. An interim final rule published April 20, 2026 moved the compliance dates to **April 26, 2027** for entities serving populations of 50,000 or more and **April 26, 2028** for smaller entities and special districts; the population is that of the city or state that runs the college, not its enrollment. Course materials, LMS content, and the tools a syllabus requires students to use are in scope. A guardrail that sends students to an inaccessible tool, or a process-evidence requirement that only works with a mouse and a screen, is an accessibility problem before it is an integrity policy.

Assistive Technology Is Now Generative

Grammar checkers, read-aloud tools, and summarizers that students receive as accommodations have added generative features. Jisc (December 2024) notes that "no AI" assessment levels did not consider assistive-technology users; George Mason disabled the generative features on its accommodation-only Grammarly licenses; Texas Tech's sample syllabus statements exempt "approved assistive technologies and approved disability accommodations" from AI restrictions; Goucher College uses a per-course AI agreement signed by the student, the instructor, and accessibility services. Your area 6 needs a path a student can use without explaining a disability to the class.

Attribution, Disclosure, Retention

Area	Published practice to borrow
Attribution	APA (2023, updated 2025), MLA (2023, updated 2025), and Chicago (18th ed.) all have AI citation formats; none treats the tool as an author. MLA puts the prompt as the title of the source.
Disclosure	The Elon and AAC&U Student Guide to AI (2025) names four elements: tools, purpose, extent (minimal, moderate, extensive), and human oversight. Loughborough requires a statement beginning "I acknowledge the use of..." with prompts and output; non-acknowledgment is misconduct. Western Michigan (March 2026) and Messiah (2025) require an AI statement on every syllabus.
Retention	Stony Brook's flagged-paper process tells students to keep time-stamped drafts and expects a revision-history conversation; Cornell's guidance is "save the chats"; QAA (2023) recommends authenticity declarations and mini-vivas and says a lack of evidence of how tools were used can itself be a misconduct question.
Integrity process	*Newby v. Adelphi* (2026) and *Harris v. Adams* (2024) point the same way from opposite outcomes: write the rule so it covers the conduct, give the student a chance to respond, and do not let a score be the finding.

Redesign Canvas: Module 6

The workbook has the full canvas (sections A to I). Use this page to draft; the Lab Manual has the viva-question commands for section G.

A. Catalog promises / legacy measures	
B. AIAS level, lane, one-sentence reason	
B. What students read on the assignment about AI	
C. Real artifact, audience, constraint	
C. Unique per student	
D. The decision, and how the rubric rewards the justification	
E. Process evidence; how kept; when reviewed	
F. Technical validation step and graded output	
G. Accountability moment; three author-only questions	

H. Rubric Rewrite (20 Points)

Legacy criterion	Redesigned criterion	Pts	Full credit looks like
Identifies suspicious activity			
Explains the attack			
Recommendations			
Report quality			

I. Cost check: instructor minutes per student, before _____ after _____. What is sampled, moved to class time, or automated locally? _____

LAB 4 · STAGE 3

Syllabus Guardrails

Two to five sentences a student can act on. Test each area against at least one scenario (numbers below) and note the result.

#	Area	Draft	Scenario, result
1	**Acceptable AI use** Which uses are allowed and which are not; how a student finds the lane for a specific assignment		
2	**Attribution** How AI contributions are credited; which style guide		
3	**Disclosure** Tools, purpose, extent, oversight; where it goes; what happens without it		
4	**Prohibited data** What may never be pasted into a tool; which environments are approved; matches the data policy		
5	**Evidence retention** What is kept, how long, in what form, when it may be requested		
6	**Accessibility** Approved assistive technology and accommodations; who decides; a private way to ask		
7	**Academic integrity** What counts as a violation; how a concern is raised; evidence relied on (no detector as sole basis); chance to respond; consequences		

Scenarios: 1 grammar tool with a rewrite button · 2 whole log pasted into a consumer chatbot · 3 disclosed AI use, cannot explain port 1433 · 4 AI-drafted firewall rule, tested and documented · 5 instructor wants to upload 34 reports to a free detector · 6 accommodation summarizer versus "no AI" · 7 four-minute transcript with no errors · 8 study group shared one chat. Full text in `lab4/student_scenarios.md`.

LAB 4 · STAGE 4 · INDIVIDUAL

The Sprint: Your Own Assignment

Fifteen minutes. Finish C and E first; they change the most. If you did not bring an assignment, use a fallback from `lab4/own_assignment_fallbacks.md`.

Assignment (course, title, points)	
A. What it claims to measure / what it actually measures	
B. AIAS level and lane	
C. Real artifact, audience, constraint, unique per student	
D. The decision with two defensible answers	
E. Process evidence and retention	
F. Technical validation	
G. Accountability moment; one author-only question	
H. The rubric row that changes most (old and new)	
I. Cost check	

> The honest test is section I. If the redesign doubles your grading time, decide now what is sampled, moved into class time, or done by a local model whose output you then verify. A redesign nobody can afford to run is the legacy assignment with extra steps.

SCORING

Lab 3 (4 Points) and Lab 4 (20 Points)

Lab 3: Threat Model

Point	Earned when	Score
1	Predictions were recorded before any model run	
2	Every false claim in the two AI outputs is listed with the command that disproves it	
3	All four submissions are scored on the legacy rubric and the three competency questions	
4	The measures-versus-claims statement names the competency gap in one sentence	

Lab 4: Redesign and Guardrails

Criterion	Points	Full credit	Partial credit	Score
Threat model carried in	3	The redesign names the gap Lab 3 found and the AI behaviors it must defeat (invented facts, missed finding, generic recommendations).	1 to 2: gap named without the behaviors, or vice versa.	
Authentic performance and decision-making	4	A real artifact for a named audience under a stated constraint; something unique per student; a decision with two defensible answers whose justification carries points.	2 to 3: artifact real but interchangeable, or decision present but the rubric rewards the answer.	
Process evidence and technical validation	4	Specific evidence named with retention terms; a validation step whose output is graded; no reliance on a detector.	2 to 3: evidence named without retention terms, or validation described but not graded.	
Accountability and rubric	3	An oral or live component with author-only questions; four rewritten rubric rows that measure what the lane can claim.	1 to 2: component present without questions, or rubric rows still reward the product alone.	
Syllabus guardrails	4	All seven areas in student-actionable language; each tested against a scenario; consistent with the interim data policy, FERPA, and the Title II rule; assistive technology addressed.	2 to 3: areas missing, untested, or one that contradicts the data policy.	
Own-assignment sprint and Evidence Log	2	Canvas sections A to H attempted for one real assignment, C and E complete; log rows complete with validation decisions.	1: sprint started, or log incomplete.	
			Total	

A Level 1 redesign (a live, supervised practical) can earn full credit if the canvas shows how it fits a 75-minute synchronous session and what happens to students who miss it. A Level 4 redesign can earn full credit only if the accountability and validation rows are strong enough to carry the grade.

APPENDIX

Printed Lab Packet

Synthetic training material. Printed copies of the range files in `/opt/nitic/session2/`. The log is the Session 1 `item5_firewall_excerpt.log`, printed in the Session 1 packet and on the range in both session folders.

The Legacy Assignment · lab3/assignment_CYB110_module6_incident_report.md

```
# CYB 110 Security Concepts, Section 02

## Module 6 Lab Assignment: Firewall Log Review and Incident Report

| | |
| --- | --- |
| Instructor | D. Okafor |
| Points | 20 |
| Due | Sunday, October 18, 2026, 11:59 p.m. (module dropbox) |
| First posted | Fall 2023; last revised August 2024 |

### Scenario

The network team has given you an excerpt from the campus perimeter firewall log for the early morning of October 13. (The log file is replaced each term; the instructions have not changed.) You are the on-duty analyst. Review the log and determine whether anything suspicious occurred.

### Instructions

1. Download `nbtc-fw01_excerpt.log` from the module page.
2. Review the log entries and identify any suspicious activity.
3. Write an incident report of 400 to 600 words that includes:
   a. A summary of what you observed
   b. Identification of the suspicious source(s) and what they were trying to do
   c. The likely goal of the attacker
   d. At least two recommended changes to the firewall rules or other defenses
4. Submit as a Word document or PDF through the module dropbox.

### Rubric (20 points)

| Criterion | 5: Excellent | 3: Satisfactory | 1: Needs work |
| --- | --- | --- | --- |
| Identifies suspicious activity | Correctly identifies the suspicious source and what it did | Identifies the source but the description is incomplete | Source not identified or incorrect |
| Explains the attack | Clear explanation of the attacker's technique and likely goal | Explanation present but vague or partly incorrect | Missing or incorrect |
| Recommendations | Two or more specific, appropriate recommendations | Recommendations present but generic | Fewer than two, or inappropriate |
| Report quality | Well organized, professional tone, 400 to 600 words, no grammar errors | Minor organization or length issues | Disorganized, off length, or many errors |

### Academic integrity

All work must be your own. Use of unauthorized aids is prohibited. See the course syllabus.

---
TRAINING MATERIAL. Northbridge Technical College, its people, and its students are fictional.
```

Submission A · submission_A_delgado.txt

Exported from the module dropbox. Formatting removed. TRAINING MATERIAL. Northbridge Technical College, its people, and its students are fictional.

Incident Report: Suspicious Network Activity on nbtc-fw01
Prepared by: Marcus Delgado, CYB 110 Section 02
Date: October 17, 2026

Executive Summary

On October 13, 2026, between approximately 02:00 and 03:20 EDT, the Northbridge Technical College perimeter firewall (nbtc-fw01) recorded a series of suspicious connection attempts originating from external IP addresses. The activity is consistent with a reconnaissance port scan followed by an attempted intrusion. This report summarizes the observed events, identifies the sources, assesses the likely objective of the attacker, and provides recommendations to strengthen the organization's overall security posture.

Observations

A review of the log excerpt reveals that the external host 203.0.113.47 initiated more than thirty connection attempts against multiple internal servers, including the web server (web01), the VPN gateway (vpn-gw), the student information system database (sis-db01), and the file server (fs01). The majority of these attempts targeted well-known service ports such as FTP (21), SSH (22), Telnet (23), SMTP (25), RDP (3389), and MySQL (3306), and were blocked by the firewall's default deny rule. Two connections to port 443 were permitted by existing allow rules, which indicates that the attacker was able to confirm that HTTPS services were exposed.

Shortly afterward, at approximately 03:02, a second external address, 198.51.100.140, made nine repeated connection attempts to the VPN gateway within roughly ninety seconds. All of these attempts failed, which strongly suggests a brute-force attack against VPN credentials.

In addition, the log shows an internal host, 192.168.1.105, communicating with the domain controller over port 88 (Kerberos) during the same window, which may indicate lateral movement following an initial compromise.

Analysis

The attacker's methodology follows the classic cyber kill chain. The initial port scan represents the reconnaissance phase, during which the adversary enumerates open services and identifies potential entry points. The subsequent VPN brute-force attempt represents the delivery and exploitation phases, in which the attacker attempts to gain initial access using guessed or stolen credentials. The attacker's likely goal is to establish a foothold on the internal network in order to exfiltrate sensitive data, such as the student records stored in the SIS database, or to deploy ransomware.

Recommendations

1. Implement rate limiting and account lockout on the VPN gateway, and enable multi-factor authentication for all remote access accounts.
2. Block 203.0.113.47 and 198.51.100.140 at the perimeter and add them to a threat intelligence blocklist.
3. Deploy an intrusion detection and prevention system (IDS/IPS) such as Snort or Suricata to alert on port-scanning behavior in real time.
4. Ensure all internal servers are fully patched and that unnecessary services are disabled to reduce the attack surface.
5. Conduct regular security awareness training for faculty and staff.

Conclusion

The events recorded on October 13 demonstrate a coordinated attempt to compromise Northbridge's network infrastructure. While the firewall successfully blocked the majority of the malicious traffic, the organization should treat this incident as a warning sign and implement the recommended controls to mitigate the risk of a successful breach in the future. Continuous monitoring and a defense-in-depth strategy remain essential to protecting the college's critical assets.

Submission B · submission_B_nair.txt

```
Exported from the module dropbox. Formatting removed. TRAINING MATERIAL. Northbridge Technical College, its people, and its students are fictional.


Priya Nair
CYB 110-02
Module 6 Incident Report

## Incident Report: Firewall Log Analysis

**Summary**

After reviewing the firewall log from nbtc-fw01 on October 13, 2026, I identified a port scan attack from the IP address 203.0.113.47. This IP address is a known malicious IP associated with botnet activity and has been reported for scanning in the past. The attacker scanned 25 ports on the Northbridge network between 2:11 AM and 2:12 AM. The firewall blocked these attempts with the default-deny rule, including an attempt to access port 443 on the web server.

**Suspicious Activity**

The attacker used a tool such as Nmap to scan common ports including FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), and database ports like MySQL (3306) and MSSQL (1433). This is called a port scan and it is used to find open ports that can be exploited. The scan targeted four servers: web01, vpn-gw, sis-db01, and fs01. Because the scan happened at 2 AM when nobody was watching, the attacker was trying to avoid detection.

**Likely Goal**

The attacker's goal was to find a vulnerable service to exploit and gain unauthorized access to the network. The attacker was probably trying to steal sensitive data such as student information or install malware. Port scanning is usually the first step in a cyber attack.

**Recommendations**

1. Block the IP address 203.0.113.47 on the firewall so it cannot scan the network again.
2. Enable an intrusion detection system (IDS) to detect and alert on port scans.
3. Update all firewall firmware and server software to the latest versions to patch vulnerabilities.
4. Train employees on cybersecurity awareness so they can recognize attacks.
5. Review the firewall rules regularly to make sure only necessary ports are open.

**Conclusion**

In conclusion, the firewall log shows a port scan attack from a malicious IP address. The firewall did its job and blocked the attack, but the college should take additional steps to improve its security. By implementing the recommendations above, Northbridge Technical College can protect its network from future attacks and keep student data safe.

The log also showed some normal internal traffic between servers, which is expected and not suspicious. Overall the firewall configuration appears to be working correctly, but continuous monitoring is important to maintain a strong security posture against evolving cyber threats in today's digital world.
```

Submission C · submission_C_ferreira.txt

```
Exported from the module dropbox. Formatting removed. TRAINING MATERIAL. Northbridge Technical College, its people, and its students are fictional.


Incident report - firewall log Oct 13
Diego Ferreira, CYB110-02

What I did

I opened the log in the lab VM and used grep to count things instead of reading all 50 lines by eye. grep -c DROP gives 25 and grep -c ACCEPT gives 25. 23 of the drops are from 203.0.113.47. It hit 23 different ports on four servers (web01, vpn-gw, sis-db01, fs01) between 02:11:04 and 02:11:48, about one port every two seconds, so this is an automated scanner, not a person. The firewall dropped all of them except two connections to 443 on web01 and vpn-gw, which are allowed because those are public services anyway. So the scan did not get anything we don't already show the whole internet.

What I think actually matters

The thing I would report first is not the scan. It's 10.30.12.118. The log only names destinations, but 10.30.12.x is the student lab subnet from the Module 4 network diagram, so this is a lab workstation. At 2:23 am it connects to web01, then SMB (445) to fs01 at 2:30, 2:46 and 2:56, then Kerberos to dc01 at 3:07, and then at 03:14:41 and 03:14:44 it tries port 1433 on sis-db01, which is the SQL port for the student information system. Those two were dropped by the deny-lab-to-admin rule. Nobody should be on a lab workstation at 3 am, and even if they were, a lab machine has no reason to talk to the SIS database. Either a student was doing something they shouldn't, or the machine is compromised and something is moving around inside the network.

Also 198.51.100.140 connected 9 times to the VPN on 443 in about 90 seconds (03:02:10 to 03:03:38). The firewall only shows that the connections were accepted. It can't tell me whether the logins worked, so I can't call it a brute force. I would check the VPN server's auth log for that address.

Recommendations

1. Find out who was logged into 10.30.12.118 between 2 and 3:15 am on Oct 13 (lab sign-in sheet, domain logon events on dc01) and image the machine if nobody was.
2. deny-lab-to-admin worked for 1433 but lab machines could still reach fs01 on 445. Restrict the lab subnet to the one share it needs, or block 445 from the lab VLAN entirely.
3. Check vpn-gw's auth logs for 198.51.100.140 and turn on lockout if it isn't on.
4. Blocking 203.0.113.47 by itself won't do much since scanners change addresses, but adding it to a blocklist is fine.

Things I'm not sure about

I don't know if 10.30.12.105 (another lab machine, 445 to fs01 at 2:43) is related or just a backup job. I also don't know what 10.20.8.40 (fs01) was doing talking to dc01 and web01, it might be normal.
```

Submission D · submission_D_kowalski.txt

```
Exported from the module dropbox. Formatting removed. TRAINING MATERIAL. Northbridge Technical College, its people, and its students are fictional.


Hannah Kowalski
CYB 110 Section 02
Module 6: Firewall Log Review and Incident Report
October 18, 2026

Summary of Observations

This report analyzes the firewall log excerpt from nbtc-fw01 covering October 13, 2026, from 02:08 to 03:14 EDT. The log contains 50 entries: 25 accepted connections and 25 dropped connections. The most prominent activity is a port scan conducted by the external IP address 203.0.113.47, which probed 23 different ports across four internal servers (web01, vpn-gw, sis-db01, and fs01) in a 44-second window beginning at 02:11:04. All 23 probes were dropped by the default-deny rule. The same address then made two accepted connections to port 443 on web01 and vpn-gw, which are publicly available services.

A second pattern involves the external address 198.51.100.140, which made nine accepted connections to the VPN gateway on port 443 between 03:02:10 and 03:03:38. In addition, an internal lab host, 10.30.12.118, made two blocked attempts to reach the SIS database server on port 1433 at 03:14, which should be reviewed.

Identification of the Suspicious Source

The primary suspicious source is 203.0.113.47. The systematic probing of ports such as FTP (21), SSH (22), Telnet (23), RDP (3389), and several database ports (1433, 1521, 3306, 5432, 27017) is characteristic of an automated reconnaissance scan intended to enumerate exposed services before an attack. The rapid, sequential timing confirms the use of a scanning tool rather than manual activity.

Likely Goal of the Attacker

The attacker's likely goal was to identify vulnerable or misconfigured services that could be exploited to gain unauthorized access to the internal network. Database ports were of particular interest, which suggests the attacker may have been seeking access to sensitive institutional data such as student records. The follow-up connections to port 443 indicate that the attacker confirmed which services were reachable.

Recommendations

1. Add 203.0.113.47 to a perimeter blocklist and subscribe the firewall to a reputation feed so similar scanners are blocked automatically.
2. Enable port-scan detection on the firewall or deploy an IDS so that rapid multi-port probes generate an alert in real time rather than being discovered during a manual log review.
3. Review the VPN gateway's authentication logs for 198.51.100.140 to determine whether the nine connections represent failed logins, and enable account lockout and multi-factor authentication if they are not already in place.
4. Investigate the lab host 10.30.12.118 to determine why it attempted to connect to the SIS database server, and consider tightening the lab subnet's access to administrative systems.

Conclusion

The firewall performed as designed by blocking the reconnaissance scan, but the log also reveals activity that warrants follow-up, including the repeated VPN connections and the internal attempt to reach the database server. Implementing the recommendations above would improve both detection and response capabilities for the college.

Note: I used ChatGPT to help me organize my report and check my grammar. The analysis and the numbers are my own work from reviewing the log.
```

Interim AI Data Use Policy · lab4/northbridge_ai_data_policy_interim.md

```
# Northbridge Technical College: Interim AI Data Use Policy

Approved by the Provost, Dr. Alicia Harmon, October 23, 2026, on the recommendation of the IT Security Office. Interim for the 2026-27 academic year pending review by the Academic Council.

## 1. Purpose

This policy states where college data may be processed by generative AI tools. It applies to all employees and to any tool, including free consumer tools, campus-hosted tools, and licensed hosted services.

## 2. Data Tiers

| Tier | Examples | Governing law |
| --- | --- | --- |
| Public | Course catalog, published schedules, press releases | None |
| Internal | Department meeting notes, budget drafts, unpublished aggregate reports | College policy |
| Confidential | Security logs, reported phishing emails, personnel matters, aggregate data that could identify individuals in small groups | College policy; contracts |
| Restricted | Education records including student work submitted for a grade, grades, rosters, accommodation letters; financial aid records; SSNs; health information | FERPA (34 CFR Part 99); GLBA Safeguards Rule (16 CFR Part 314); HIPAA where applicable |

## 3. Approved Environments

| Environment | Public | Internal | Confidential | Restricted |
| --- | --- | --- | --- | --- |
| Consumer AI tools (free or personal accounts) | Yes | No | No | No |
| Licensed hosted service (enterprise agreement, no training on inputs, 30-day retention) | Yes | Yes | Yes, with approval of the data owner | No |
| Northbridge AI Server (`ai.northbridge-tc.example`, on campus, Ollama and Open WebUI) | Yes | Yes | Yes | Yes, de-identified where the task allows, with chat history disabled for Restricted data |

## 4. Requirements

1. Restricted data never enters a consumer tool or any service without a college contract. Uploading student work to a third-party website, including AI-detection services, is processing of an education record and requires a contract that meets the FERPA school-official conditions.
2. Any AI output used in a decision about a person (grading, discipline, hiring, aid) must be reviewed by a named employee who records the review.
3. Employees record consequential AI use in a unit log (prompt, model, output, decision).
4. Suspected violations go to the Information Security Office (555-0170) and are handled under the Data Incident Procedure, not as misconduct, during the interim period.

## 5. Scope Note

This policy governs how data is handled. It does not decide whether students may use AI in coursework or how faculty assess student learning. Those questions belong in course syllabi and department assessment standards.

---
TRAINING MATERIAL. Northbridge Technical College, its people, and its students are fictional.
```

5:25

Exit Ticket

Submit through the form link in chat, or answer here and photograph it.

1. Which of the seven guardrail areas are you least sure how to write for your own course, and why?

2. Which assignment will you redesign first, and which of the five properties does it most need?

3. On a scale of 1 to 5, how ready are you to run Lab 3 or Lab 4 with your own students?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 What would move you up one point?

> **Session 3 preview.** The invented IP address in Module 6 is the problem retrieval exists to solve. Next session Northbridge's security office grounds its models in the college's own documentation and current threat intelligence, and the mission tests whether the answers get better or just more confident.